Chapter 15 Review

Campbell Biology, AP Edition, 12e

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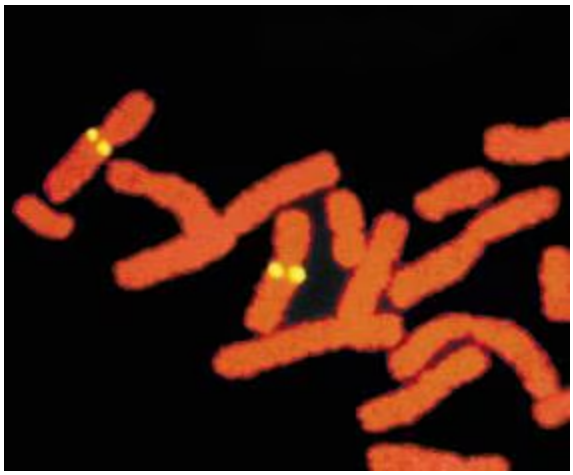


Summary of Key Concepts

What observed action of chromosomes made them the likely location of the genetic factors Mendel postulated as carriers of genetic information? **(BIG IDEA 3)**

Concept 15.1: Morgan showed that Mendelian inheritance has its physical basis in the behavior of chromosomes

- Morgan's work with an eye color gene in *Drosophila* led to the **chromosome theory of inheritance**, which states that genes are located on chromosomes and that the behavior of chromosomes during meiosis accounts for Mendel's laws.

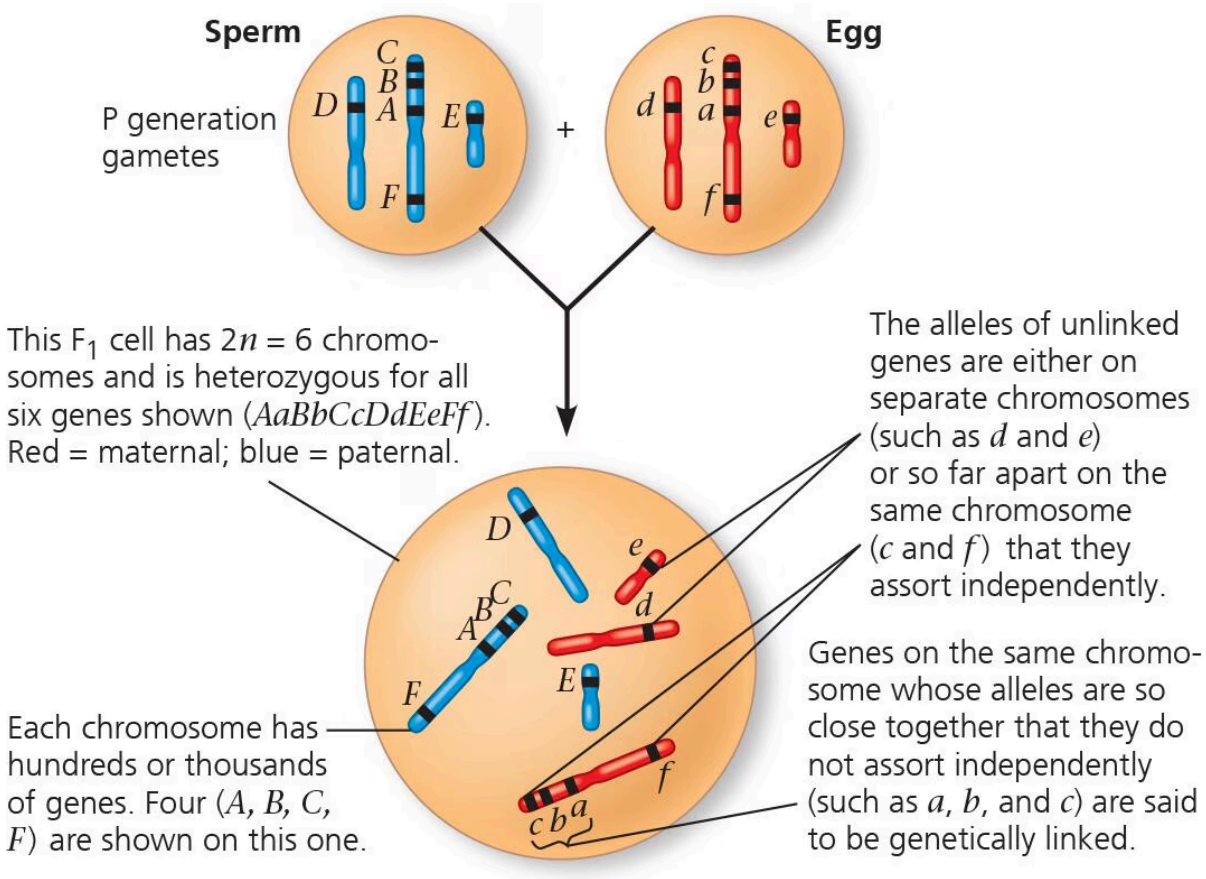
What characteristic of the sex chromosomes allowed Morgan to correlate their behavior with that of the alleles of the eye color gene?

Concept 15.2: Sex-linked genes exhibit unique patterns of inheritance

- Sex is often chromosomally based. Humans and other mammals have an X-Y system in which sex is largely determined by whether a Y chromosome is present. Other systems are found in birds, fishes, and insects.
- The sex chromosomes carry **sex-linked genes**, virtually all of which are on the X chromosome (**X-linked**). Any male who inherits a recessive X-linked allele (from his female parent) will express the trait, such as color blindness.
- In mammalian females, one of the two X chromosomes in each cell is randomly inactivated during early embryonic development, becoming highly condensed into a **Barr body**.

Why are males affected by X-linked disorders much more often than females?

Concept 15.3: Linked genes tend to be inherited together because they are located near each other on the same chromosome



P generation gametes:

Sperm: three blue chromosomes with six genes shown (A B C D E F). Chromosome 1 has uppercase D, chromosome 2 has uppercase C B A close together and uppercase F at the opposite end, and chromosome 3 has uppercase E.

Egg: three red chromosomes. Chromosome 1 has lowercase d. Chromosome 2 has lowercase cba all close together, and lowercase f at the opposite end. Chromosome 3 has lowercase e.

Arrows indicate fertilization between the parental gametes.

F 1 cell: This F 1 cell has 2n equals to 6 chromosomes and is heterozygous for all six genes shown (upper A lower a, upper B lower b, upper C lower c, upper D lower d, upper E lower e, upper F lower f). Red = maternal; blue = paternal. Chromosomes are:

- Blue: C B E all uppercase close together, F uppercase at the opposite end.
- Blue: with uppercase D.
- Red: lowercase c b a all close together at one end with lowercase f at the opposite end.

- Blue: with uppercase E.
- Red: with lowercase d.
- Red: with lowercase e.

Each chromosome has hundreds or thousands of genes. The alleles of unlinked genes are either on separate chromosomes or so far apart on the same chromosome that they assort independently. Genes on the same chromosome whose alleles are so close together that they do not assort independently are said to be genetically linked.

- An F1 dihybrid testcross yields **parental types** with the same combination of traits as those in the P generation parents and **recombinant types** (**recombinants**) with new combinations of traits not seen in either P generation parent. Because of the independent assortment of chromosomes, unlinked genes exhibit a 50% frequency of recombination in the gametes. For genetically **linked genes, crossing over** between nonsister chromatids during meiosis I accounts for the observed recombinants, always less than 50%.
- The order of genes on a chromosome and the relative distances between them can be deduced from recombination frequencies observed in genetic crosses. These data allow construction of a **linkage map** (a type of **genetic map**). The farther apart genes are, the more likely their allele combinations will be recombined during crossing over.

Why are specific alleles of two distant genes more likely to show recombination than those of two closer genes?

Concept 15.4: Alterations of chromosome number or structure cause some genetic disorders

- **Aneuploidy**, an abnormal chromosome number, can result from **nondisjunction** during meiosis. When a normal gamete unites with one containing two copies or no copies of a particular chromosome, the resulting zygote and its descendant cells either have one extra copy of that chromosome (**trisomy**, 2n+1) or are missing a copy (**monosomy**, 2n−1). ****Polyploidy** (extra sets of chromosomes) can result from nondisjunction of all chromosomes.
- Chromosome breakage can result in alterations of chromosome structure: **deletions, duplications, inversions**, and **translocations**. Translocations can be reciprocal or nonreciprocal.
- Changes in the number of chromosomes per cell or in the structure of individual chromosomes can affect the phenotype and, in some cases, lead to disorders. Such alterations cause **Down syndrome** (usually due to trisomy of chromosome 21), certain cancers associated with chromosomal translocations that occur during mitosis, and various other human disorders.

Why are inversions and reciprocal translocations less likely to be lethal than are aneuploidy, duplications, deletions, and nonreciprocal translocations?

Concept 15.5: Some inheritance patterns are exceptions to standard Mendelian inheritance

- In mammals, the phenotypic effects of a small number of particular genes depend on which allele is inherited from each parent, a phenomenon called **genomic imprinting**. Imprints are formed during gamete production, with the result that one allele (either maternal or paternal) is not expressed in offspring.
- The inheritance of traits controlled by the genes present in mitochondria and plastids depends solely on the maternal parent because the zygote's cytoplasm containing these organelles comes from the egg. Some diseases affecting the nervous and muscular systems are caused by defects in mitochondrial genes that prevent cells from making enough ATP.

Explain how genomic imprinting and inheritance of mitochondrial and chloroplast DNA are exceptions to standard Mendelian inheritance.